ARSENIC DETECTION SYSTEM

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The Problem

- Detecting arsenic contaminations
- A toxic chemical
  - can cause nausea, cancer, gastrointestinal disorders and other health problems (Sadee et al, 2025)
- Tools for event detection:
  - Flow sensors
  - Chlorine sensors

Our Approach

1. Simulating real scenarios via the EpytFlow API
    a. constant chlorine injection pattern
    b. setting up chlorine and flow sensors for every node

2. Collecting the data of 20 scenarios to build an arsenic contamination dataset

(Zhao et al, 2014)

Our Approach

3. Defining the evaluation metrics:
   a. Accuracy: $\frac{\#\,correct\ prediction}{\#\,tot\ predictions}$
   b. Probability of Detection: $\frac{TP}{TP+FN}$
   c. False Alarm Rate: $\frac{FP}{FP+TN}$
   d. Detection Latency Time: counted from the time when the peak of the event has taken place (Raciti et al. 2012)

(Zhao et al, 2014)

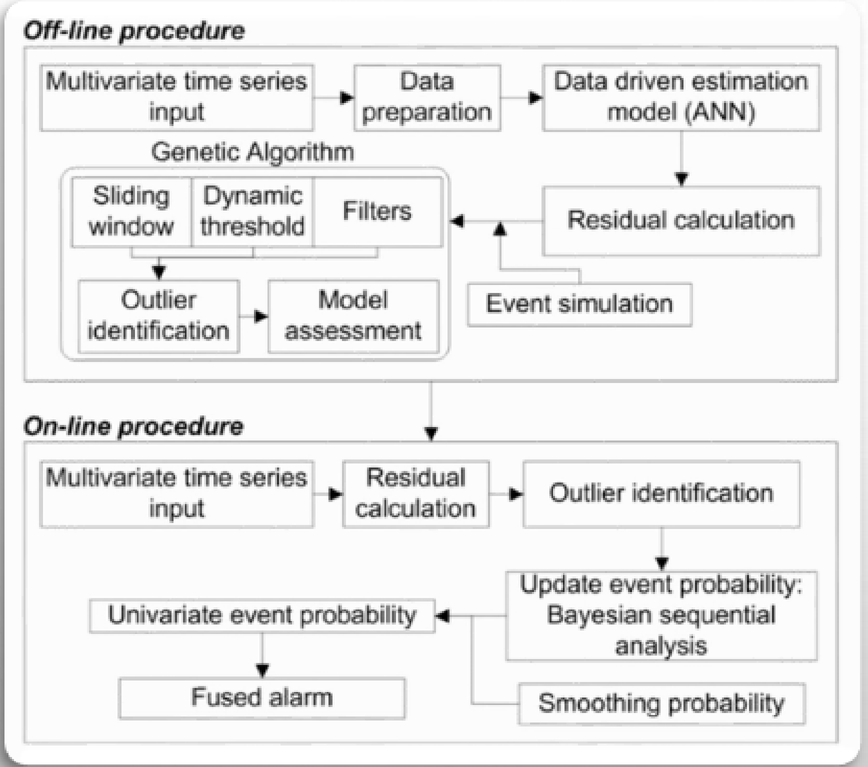
Our Approach

4. Comparing two anomaly detection methods
    a. Artificial Neural Network (ANN) (Arad et al, 2013)
    b. Linear Prediction (McKenna et al, 2008)

5. Optimizing the location of chlorine sensors for a better anomaly detection
    a. Greedy Randomized Adaptive Search Procedure (Murray et al, 2009)

(Zhao et al, 2014)

Artificial Neural Network



Main idea: use of the Genetic Algorithm to update the dynamic thresholds parameters:

1. Sliding Window Size
2. Positive Dynamic Thresholds (PDT)
3. Negative Dynamic Thresholds (NDT)
4. Positive Filters (PF)
5. Negative Filters (NF)

(Arad et al, 2013)

Linear Prediction

- $Z^*(t) = \sum_{i=1}^{P} a_i Z(P-1)$

- Anomaly score:
  - $\delta(t) = Z^*(t) - Z(t)$

(McKenna et al., 2008)

Greedy Randomized Adaptive Search Procedure

- P-median problem (find best locations in a network)
- Find best sensor locations by gradually shifting the sensors
- Look whether shift improved predictions' performance
- If so:
  - Keep the shift and keep shifting
- If not:
  - Don't keep the shift
  - Shift in another direction/eventually stop shifting (best positions found)

(Murray et al., 2010)

Current Results

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Absolute frequencies:
              detected as positive  detected as negative
is positive                   3535                   227
is negative                  47754                  2862


Relative frequencies (in percent):
              detected as positive  detected as negative
is positive                   6.50                  0.42
is negative                  87.82                  5.26


Conditional frequencies for actual arsenic contamination (in percent):
                       is positive
Detected as positive         93.97
Detected as negative          6.03


Conditional frequencies for no arsenic contamination (in percent):
                       is negative
Detected as positive         94.35
Detected as negative          5.65
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- Linear Prediction of Chlorine values
  - Regression score: 0.81 => In most cases, correct chlorine values predicted
  - No overfitting concerning anomalies
  - Massive overprediction of contamination

THANK YOU FOR LISTENING!

Questions?

References

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